

Project (CN6000)

Initial Proposal Form

Programme: BSc (Hons) Data Science and AI

Year: 2025

Student Number: u2611208

Title: The Impact and Predictive Power of On-Chain Data on Cryptocurrency Valuation.

Aim: Design & implement a deep learning model that utilises comprehensive on-chain and technical data to understand its impact on the short-term price movements of a major cryptocurrency.

Objectives:

- Review what has been done within this field before to establish a clear development and implementation plan.
- Research the existing techniques along with primary metrics and indicators that serve as primary indicators for the model.
- Collect essential on-chain data from across sources to maintain integrity and ensure synchronization.
- Clean the collected data to ensure it correctly reflects our needs.
- Implement a deep learning architecture designed to multivariate time-series forecasting.
- Train, tune and validate the model using an optimal feature set identified from the research conducted before or while training.
- Evaluate the accuracy of the model, using computational metrics against the traditional price-only models.
- Reflect on the findings to better understand the significance and impact on-chain data has on cryptocurrency price.

Rationale:

The cryptocurrency market is notorious for its volatility, with current valuation models relying heavily on speculative technical analysis rather than fundamental economics. Unlike traditional assets, crypto offers transparent, public, and immutable on-chain data reflecting actual network usage and investor behaviour. This project is necessary to bridge the gap between speculative modelling and data-driven fundamental analysis. By utilizing a deep learning approach like LSTM, which excels at recognizing temporal dependencies in complex datasets, I aim to empirically demonstrate which on-chain metrics (such as investor profitability ratios and liquidity flows) possess the highest predictive power. The findings will provide a scientific basis for investment strategies, moving beyond simple price charts to leverage the unique data advantages inherent in blockchain technology.

Facilities Required:

- Python language for development including libraries like NumPy, Pandas, Scikit-learn, TensorFlow/Keras or PyTorch and Matplotlib/Seaborn .
- Computational power to handle the model.
- Outlets to source credible data for training.

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